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Interpretation of an Interwell Partitioning Tracer Test in a Multi-Layer Carbonate Reservoir

Samuel Fontalvo and Maxim Yutkin, King Abdullah University of Science and Technology, Saudi Arabia; Majid Hassanizadeh, SimTech, Stuttgart University, Germany / Utrecht University, The Netherlands; Tadeusz Patzek, King Abdullah University of Science and Technology, Saudi Arabia

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Abstract

Interwell Partitioning Tracer Test (IPTT) estimates remaining oil for economic evaluation of reservoir operations. It involves waterflooding and co-injection of two chemicals: one water soluble (conservative) and another that also partitions into the oil phase. The downstream concentration of injected chemicals yields concentration history (CH). Cooke's interpretation of CH estimates remaining oil from tracer arrival times. Complex reservoirs need extra interpretation effort. This study reanalyzes IPTT data from a giant Saudi Arabian reservoir, as published by Sanni et al. (2016) [1].

We introduce the following reservoir features that were overlooked in the original study: multi-layer structure, mass transfer resistances, stagnant zones, and physical properties continuity. We combine these features in a physically reasonable way to capture the first-order effects. Namely, we add extra terms to the well-known advection-dispersion equation (ADE) for each feature and use separate ADE for each reservoir layer preserving physical properties continuity in each layer. Consequently, partitioning tracer transport is described by an ADE with mass-transfer resistance to the stagnant oil phase in each layer. The oil phase is continuous and allows for partitioning tracer communication between the layers. The conservative tracer follows classic ADE in both layers. The system of differential equations is solved numerically with COMSOL.

The key results from our analysis include improved mass conservation and overlooked remaining oil in the second layer. Specifically, our model accounts for all recovered mass for both conservative and partitioning tracers. More importantly, our model allows for estimation of remaining oil in the second layer. Remaining oil saturation in the first layer is 17% — close but slightly less than in the original study. The second layer hosts 25% of remaining oil.

To the best of our knowledge, this is the first published result that interprets IPTT data for a multi-layer reservoir with stagnant zones, inter-layer oil communication, and slow partitioning tracer diffusion in oil phase. Our analysis revealed hydrocarbons in a second layer, missed in the original report. This study also demonstrates how proper physics-based test interpretation can provide guidance on essential data collection to improve IPTT analyses.

Introduction

The development of modern society, current energy portfolio, and any future adjustments of this portfolio and inextricably linked to fossil fuels. Consequently, the (efficient) recovery of these resources through capitalization on the existing technologies and new product development remain crucial for the future of many industry and civil sectors that rely on inexpensive energy streams. One technological necessity is the determination of remaining oil in a reservoir to inform decisions regarding the implementation of future field development programs. There are two primary methodologies to determine remaining oil saturation: single well and interwell partitioning tracer test. Single well tests are fast and inexpensive, but probe only near-wellbore region. Interwell tests require longer times and are more expensive; however, they probe larger reservoir volumes. An interwell partitioning tracer test (IPTT) involves the injection of at least two tracers in the injector well: one conservative and one partitioning. During the test these tracers are transported from the injector to the producer, and brine samples are collected at the producer well and plotted against time resulting in a concentration history (CH) curve. The conservative tracer is inert and does not undergo any change in the reservoir while transported by the aqueous phase. The partitioning tracer is miscible with oil and thus can be transported by the aqueous phase and diffuse through the oleic phase.

Interpretation of IPTT involves fitting tracer transport models with underlying physics assumptions to the CH data. The simplest model is the advective-dispersive equation (ADE) for conservative tracer transport. This model implies the tracer is transported only by advection and hydrodynamic dispersion. Such model requires proper choice of boundary conditions, reservoir and transport physics. While there are many choices of boundary conditions (BC), Robin BC at the inlet and Neumann BC at the outlet (i.e., Danckwerts-type) are known to conserve mass at high longitudinal dispersion coefficients. This is especially critical, as both laboratory and field tracer data from limestone formations has been shown to have high dispersion coefficients due to heterogeneities. However, it is not uncommon to find alternative BC combinations for tracer test interpretation (Welty & Gelhar 1994[2], Viig et al. 2013[3], Huseby et al. 2015[4], Hartvig et al. 2015[5], Sanni et al. 2015[6], Sanni et al. 2016[1], Sanni et al. 2017[7]).

Estimation of remaining oil saturation requires also the solution of the partitioning tracer transport. The simplest model for the partitioning tracer transport uses a retardation factor to account for the delayed flow and arrival time of partitioning tracer depending on oil saturation. Cooke in 1971[8] estimated the oil saturation using the difference in arrival times of the conservative and partitioning tracers, and the partitioning equilibrium coefficient. Cooke's approach assumes equilibrium concentration of partitioning tracer in brine and oil. Historically, such an approach has been used in many IPTTs interpretation (Welty & Gelhar 1994[2], Viig et al. 2013[3], Huseby et al. 2015[4], Hartvig et al. 2015[5], Sanni et al. 2015[6], Sanni et al. 2016[1], Sanni et al. 2017[7]). However, equilibrium partitioning assumptions may not explain the long spread of tracer data in some IPTT. Cooke's interpretation of IPTT also involves selection of the peak of the concentration history as arrival time of the tracers. Cooke's method simplicity brings also limitations, as pointed out by Tang and Harker[9], [10] and Fontalvo et al.[11]. For example, the method assumes equilibrium conditions and cannot be applied when non-equilibrium conditions are present. Another issue in the original formulation is the reliance on tracer concentration peak(s) for measurement of breakthrough times. As shown by Fontalvo et al.[11], such an approach may not be valid in all cases, particularly when tracer flow is dominated by mass-transfer or diffusion into stagnant zones or isolated oil patches. Recently, we have introduced a model with non-equilibrium behavior the partitioning tracer that arises from unswept oil blobs, which is especially relevant for heterogeneous formations (Fontalvo et al. 2025[11]). We analyzed an IPTT from a giant carbonate field, based on Sanni et al. (2016)[1], where non-equilibrium partitioning tracer model captures the behavior of the partitioning tracer, better than the equilibrium partitioning tracer model. We also demonstrated that the arrival time of a tracer corresponds to the mean residence time and not the peak time. Although the non-equilibrium partitioning tracer model of Fontalvo et al. 2025[11] captures well most of the partitioning tracers data, it underperforms at the very late times of the test, and does not account for all the mass of the partitioning tracers. These issues signal that the non-equilibrium model alone does not explain the partitioning tracer transport.

In this study, we address the partitioning tracer mass conservation issues from Fontalvo et al. 2025[11]. We introduce an important feature that is missing in our re-analysis of Sanni et al. 2016 data. Namely, we introduce mass communication of partitioning tracer between the reservoir layers through remaining oil. We define two layers: one fast and one slow. We assume that while the conservative tracer is transported separately in each layer, the partitioning tracer transport involves the communication of the partitioning tracers between the layers. The interlayer communication is explained by the existence of two types of oil saturation: microscopic and macroscopic. Microscopic oil saturation refers to the interstitial capillary-trapped oil in pores. Microscopic oil saturation occurs in the well flooded zones. Macroscopic oil saturation refers to oil saturation because of the oil clusters that exist in the REV scale. Macroscopic oil saturation arises from the unswept big oil blobs after waterflooding and is mostly responsible for the non-equilibrium partitioning conditions. The interlayer oil communication improves the mass balance conversion of the partitioning tracer and enables estimation of the remaining oil in the second layer.

Methodology

Physical model

Mass conservation equations for tracer transport.

Conservative tracer

The one-dimensional macroscopic transport for a conservative tracer traversing in the aqueous phase through a homogeneous porous medium in the presence of a stagnant oil phase is described by

$$\frac{\partial C_{w,i}}{\partial t} + v_i \frac{\partial C_{w,i}}{\partial x} - D_{H,i} \frac{\partial^2 C_{w,i}}{\partial x^2} = 0 \tag{3}$$

where i denotes the layer number ($i = 1, 2$), $C_{w,i}$ and $C_{o,i}$ are average tracer concentrations in the water and oil phases of layer i [$M\,L^{-3}$], respectively, v_i is the interstitial velocity of layer i [$L\,T^{-1}$], $D_{H,i}$ is the hydrodynamic dispersion coefficient in the longitudinal direction of layer i [$L^2\,T^{-1}$], ϕ_i is the fluid-accessible porosity [-]. The interstitial velocity, v_i, is related to Darcy velocity, u_i, by $v_i = u_i/\phi_i S_{w,i}$ where $S_{w,i}$ is the saturation of water of layer i [-].

In dimensionless form, Eq. 3 becomes

$$\frac{\partial \tilde{C}_{w,i}}{\partial \tilde{t}} + \frac{\partial \tilde{C}_{w,i}}{\partial \tilde{x}} - \frac{1}{Pe_{L,i}} \frac{\partial^2 \tilde{C}_{w,i}}{\partial \tilde{x}^2} = 0 \tag{4}$$

where $Pe_{L,i} = v_i L/D_{H,i}$ is the longitudinal Péclet number of layer i, $\tilde{x} = x/L$, $\tilde{t} = v_i t/L$, L is the characteristic length between inlet and outlet (injector and producer), and $\tilde{C}_{w,i}(\tilde{x}, \tilde{t}) = \left(C_w(x, t) - C_{init}\right)/\left(C_{inj} - C_{init}\right)$, where subscripts "init" and "inj" denote initial and injected concentrations, respectively. For the case where there is zero concentration in the reservoir at initial time, $\tilde{C}_{w,i}(\tilde{x}, \tilde{t}) = C_w(x,t)/C_{inj}$.

Partitioning tracer

For a partitioning tracer, two approaches are possible. If the rate of mass transfer of partitioning tracer withing the oil phase is fast in comparison to advective flux, one may assume equilibrium partitioning. In cases where the mass transfer rate is slow relative to the advective flux, the local equilibrium assumption is no longer valid. We describe the equilibrium and non-equilibrium partitioning tracer model in Fontalvo et al. 2025 [11].

In this work, we extend the non-equilibrium partitioning model for the two (fast and slow) layers, with microscopic oil saturations for each layer, and a macroscopic oil saturation which allows communication between the layers. This model combines diffusion mass-transfer of the partitioning tracer in the microscopic trapped oil in layer i, $\alpha_{m,i}(KC_{w,i} - C_{o,i})$, plus the mass-transfer by diffusion of the partitioning tracer in the macroscopic oil region, $\alpha_{M,i}(KC_{w,i} - C_{o,M})$. The macroscopic oil zone is given by a baffle layer which is oil saturated and allows communication between the two layers. The resulting equations read as follows

$$\phi_i S_{w,i}\frac{\partial C_{w,i}}{\partial t} + u_i\frac{\partial C_{w,i}}{\partial x} - \phi_i S_{w,i} D_{H,i}\frac{\partial^2 C_{w,i}}{\partial x^2} = -\phi_i \alpha_{m,i}\left(KC_{w,i} - C_{o,i}\right) - \phi_i \alpha_M\left(KC_{w,i} - C_{o,M}\right) \tag{5}$$

and

$$\phi_i S_{o,i}\frac{\partial C_{o,i}}{\partial t} = \phi_i \alpha_{m,i}\left(KC_{w,i} - C_{o,i}\right) \tag{6}$$

and

$$\frac{\partial C_{o,M}}{\partial t} = \alpha_M \sum_{i=1}^{N}\left(KC_{w,i} - C_{o,M}\right) \tag{7}$$

where i denotes the layer number (i = 1, 2), N is the number of layers (N = 2), $C_{o,M}$ is the average tracer concentration in the macroscopic oil phase [$M\,L^{-3}$], $\alpha_{m,i} = a_V^i D_m^o / \delta$ is the effective microscopic mass transfer rate coefficient for tracer oil-phase diffusion of layer i [T^{-1}], and α_M is the effective macroscopic mass transfer rate coefficient for tracer oil-phase diffusion in the macroscopic oil zone [T^{-1}], $K = C_o/C_w$ is the equilibrium partitioning coefficient [-], a_V^i is the specific surface area of the oil-brine interface [L^{-1}]; D_m^o is the partitioning tracer molecular diffusion coefficient in the oil [$L^2\,T^{-1}$]; and δ is the average transverse characteristic dimension of oil blobs [L] . The final system of equations includes five equations: Eq. 5 for the aqueous phase in each layer, Eq. 6 oleic phase in each layer, and Eq. 7 for the mass-transfer of the macroscopic oil zone.

In dimensionless form, Eq. 5 for the mobile phase for each layer is defined as

$$\frac{\partial \tilde{C}_{w,i}}{\partial \tilde{t}} + \frac{\partial \tilde{C}_{w,i}}{\partial \tilde{x}} - \frac{1}{Pe_{L,i}}\frac{\partial^2 \tilde{C}_{w,i}}{\partial \tilde{x}^2} = -\frac{Da_{m,i}}{S_{w,i}}\left(K\tilde{C}_{w,i} - \tilde{C}_{o,i}\right) - \frac{Da_M}{S_{w,i}}\left(K\tilde{C}_{w,i} - \tilde{C}_{o,M}\right) \tag{8}$$

Then, Eq. 6 for the immobile microscopic oil phase becomes

$$\frac{\partial \tilde{C}_{o,i}}{\partial \tilde{t}} = \frac{Da_{m,i}}{S_{o,i}}\left(K\tilde{C}_{w,i} - \tilde{C}_{o,i}\right) \tag{9}$$

and, Eq. 7 for the immobile macroscopic oil zone is

$$\frac{\partial \tilde{C}_{o,M}}{\partial \tilde{t}} = \frac{Da_M}{S_{o,M}}\sum_{j=1}^{N}\left(K\tilde{C}_{w,i} - \tilde{C}_{o,M}\right) \tag{10}$$

where $Da_{m,i} = a_V^i D_m^o L / \delta v_i$ is the microscopic mass-transfer Damköhler number of layer i, and Da_M is the macroscopic mass-transfer Damköhler number.

Boundary conditions. The solution of these partial non-homogeneous differential equations requires two boundary conditions (BC) at the inlet and outlet for $C_{w,i}$, and two initial conditions for $C_{w,i}$ and $C_{o,i}$. The initial conditions (t = 0) are $C_{w,i}(x, 0) = 0$ and $C_{o,i}(x, 0) = 0$. Mass conservation for highly dispersive formations requires the use of Robin BC at the inlet and Neumann BC at the outlet. The Robin/Neumann BC were formulated by Danckwerts (1953)[12]. The injected concentration is of the pulse injection type, where t_s is the injected slug/pulse duration.

Model fitting. The fitting of field IPTT data is described in detail by Fontalvo et al. 2025 [11].

Results

We start our analysis by presenting the original study by Sanni et al. (2016)[1] and the re-analysis of this study by Fontalvo et al. (2025)[11] that includes the equilibrium and non-equilibrium partitioning models. Next, we present the dual-layer mass-conservative model with partitioning tracer transport between the layers.

The interwell partitioning tracer test was conducted after a mature waterflooding of a giant Jurassic-age carbonate reservoir (Sanni et al. [1], [6], [7]). The reservoir consists of several layers. One layer consists of diagenetic dolomites, referred to here as the first layer. The other layers are lumped into one layer and referred to here as the slow layer. The IPTT configuration comprises an injector and a producer well, separated by approximately 610 m. The IPTT involved the injection of one conservative tracer, WT-60, and three partitioning tracers, WTP-2 (K = 2.1), WTP-3 (K = 2.1), and WTP-4 (K = 4). Upon closer examination, the published datasets appear to originate from different experiments. We use the data presented by Sanni et al. (2016)[1].

The original approach in Sanni et al.[1] uses an analytic solution proposed by Viig et al. (2013)[3]. Total mass injected, dispersion coefficient, and pulse time are the three fitting parameters of Viig's model. However, Viig's analytic solution applies non-mass conservative boundary conditions. In other words, it does not conserve mass at the longitudinal Péclet numbers realized in the reservoir. In contrast, Fontalvo et al. (2025)[11] use a mass-conservative analytic solution for conservative tracer transport with Danckwerts—type boundary conditions (Robin BC at the inlet and Neumann BC at the outlet). Here we follow approach of Fontalvo et al. (2025)[11].

Figure 1 presents conservative tracer concentration histories, C(t) (solid black circles), converted from digitized residence time distributions, E(t),[1] using the mass injected and the average injection rate obtained from Sanni et al. (2015)[6]. The solid black line in Figure 1 represents the best fit sum for the conservative tracer concentration histories (dashed and dotted lines for first and second layers, respectively) of two analytic solutions for Eq. 3 with Robin/Neumann boundary conditions. The tracer concentration data shows two peaks at approximately 240 and 830 days, from the fast and slow layers, respectively. Calculated mean residence times are 380 and 870 days, for the fast and slow layers, respectively.

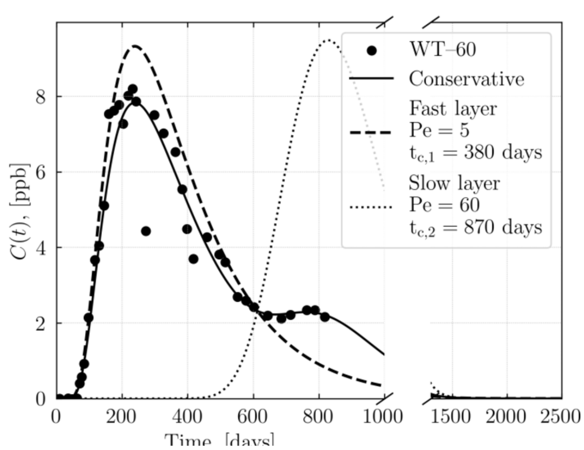

Figure 1—Fitted sum concentration history for the conservative tracer from two layers of the reservoir. Solid black circles represent the field conservative tracer (WT-60) data. The lines correspond to the fitted conservative tracer using an analytic solution to Eq. 3 with Robin—Neumann boundary conditions. The dashed line represents the fast layer, and the dotted line corresponds to the slow layer. The solid line is the volume weighted sum of each layer's contribution.

The volumetric flow distribution is 0.85 for the first layer and 0.15 for the second layer. The volumetric flow distribution follows an optimization problem constrained by the relationship between the injected pulse time of the first layer ($t_{s,1}$) and the injected pulse time of the second layer ($t_{s,2}$). $t_{s,1} = ft_{s,2}$ are related by a factor, f, that depends on the ratio of velocity between the layers ($f = v_1/v_2$).

Based on the conservative tracer fit results, we conclude that the mean interstitial fluid velocity in the second layer ($v_2 = 0.7$ m/day) is approximately 2.3 times slower than in the first layer ($v_1 = 1.6$ m/day). The hydrodynamic dispersion in the first layer is $2.27\times10^{-3}\ m^2/s$ ($Pe_{L,1} = 5$), while it is $8.25\times10^{-5} m^2/s$ in the second layer ($Pe_{L,2} = 60$). Similar or higher values of dispersion coefficients have been found in the literature[14], [15]. Finally, the total recovered mass of the conservative tracer is 46%. Mass balance demands that the recovered mass fraction of partitioning tracer is the same.

Figure 2 presents partitioning tracer concentration history curves, C(t), converted from digitized residence time distributions, E(t),[1] using the mass injected and the average injection rate obtained from Sanni et al. (2015)[6]. Three partitioning tracers were injected along with conservative tracer: WTP-2 (K = 2.1) and WTP-3 (K = 2.1) represented by blue solid circles and blue crosses, respectively, and WTP-4 (K = 4) represented by red solid circles. Figure 2 shows four partitioning tracer transport models.

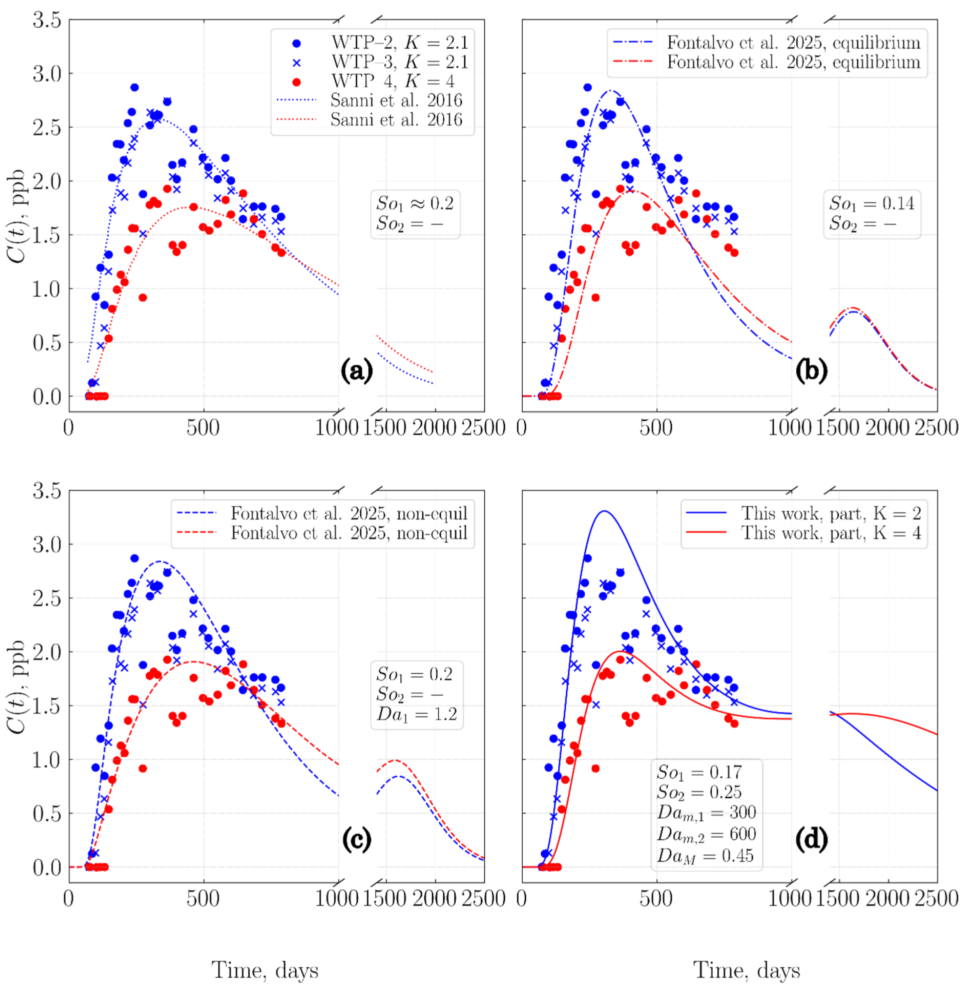

Figure 2—Concentration history curves for the partitioning tracers. The blue filled circles and crosses represent the partitioning tracers WTP-2 and WTP-3, both with K = 2.1. The red filled circles represent the partitioning tracer WTP-4 with K = 4. The partitioning tracer models are (a) Sanni et al. (2016) model represented by blue and red dotted lines which fits the tracers with K = 2.1 and K = 4, respectively; (b) Fontalvo et al. (2025) equilibrium model represented by blue and red dotted-dashed lines for tracer models with K = 2 and K = 4, respectively; (c) Fontalvo et al. (2025) non-equilibrium model represented by blue and red dashed lines for tracer models with K = 2 and K = 4, respectively; (d) partitioning tracer model with two layers, assumes equilibrium conditions for the microscopic oil saturation and non-equilibrium conditions for the macroscopic oil saturation. This model is explained by Eqs. 5-7. The blue and red solid lines correspond to tracer models with K = 2.1 and K = 4, respectively.

Figure 2(a) presents partitioning tracer field data with the original model by Sanni et al. 2016[1]. The blue dotted line corresponds to the fitting to the tracer WTP-2 (K = 2.1) and the red dotted line corresponds to the fitting to the tracer WTP-4 (K = 4). Conservative and partitioning tracer models were fit to the data independently, each yielding distinct dispersivity characteristics of the porous media and residence times. The residence times were used to calculate the remaining oil saturation. While independent curve fitting is a viable approach, it disregards the continuity of physical properties between conservative and partitioning tracer flows. Partitioning tracer mass recovery was between 26-29%[1]. The remaining 19-16% of mass is unaccounted and likely trapped in the second layer. The estimate of oil saturation found by Sanni et al. 2016 was ~0.2.

In contrast, Figure 2(b) presents the equilibrium partitioning tracer transport model for the first layer with preserved continuity of physical properties from Fontalvo et al. (2025)[11]. The blue and red dot-dashed lines correspond to the fitting of the equilibrium partitioning tracer transport model with K = 2 and K = 4, respectively. The equilibrium partitioning tracer transport model captures only the early arrival times of the partitioning tracers, while dropping fast after the peak, thus not capturing the late times. The remaining oil saturation found with the equilibrium model is 0.14. Additionally, the partitioning tracer mass recovery is between 26-29%[11] for the fast layer, significantly less than the conservative tracer mass recovery of ~46% for both layers. The conservative tracer mass recovered from the second layer is 7%. Hence, we expect 7% of the partitioning tracer to be transported in the slow layer. This indicates that ~13% of the mass is still missing and it is not explained by the equilibrium partitioning tracer model. Therefore, the equilibrium partitioning tracer model fails to correctly capture the physics of the reservoir.

Figure 2(c) shows the non-equilibrium partitioning tracer transport model for the fast layer with preserved continuity of physical properties from Fontalvo et al. (2025)[11]. The blue and red dashed lines correspond to the fitting of the equilibrium partitioning tracer transport model with K = 2 and K = 4, respectively. The non-equilibrium model captures the early times and highly skewed spread of the data after the peak (late times) of the partitioning tracers. The remaining oil saturation estimation found by the non-equilibrium model is 0.2, which is consistent with the cross-validated oil saturation estimates with other methods (Sanni et al. (2016)[1]). Although the non-equilibrium partitioning tracer model captures most of the partitioning tracer data, towards the end of the collected data the models do not follow the behavior of the partitioning tracers. The mass recovered from the partitioning tracer model from the fast layer is ~34%, still less than the mass recovered from the conservative tracer ~46% for both layers. The conservative tracer mass recovered from the second layer is 7%. Hence, we expect 7% of partitioning tracer to be transported in the slow layer. This indicates that ~5% of the mass is still missing and it is not explained by the non-equilibrium partitioning tracer model. These issues signal that the non-equilibrium model alone does not explain the partitioning tracer transport. Moreover, these models only give conclusions about the first layer, and the second layer oil saturation is unknown.

Figure 2(d) shows the fitting of the partitioning tracer transport model for a dual-layer system with microscopic and macroscopic oil saturation zones. It uses the non-equilibrium partitioning approach, extended for two layers, and distinguishes between microscopic and macroscopic oil saturated zones, as described by Equations 5 – 7. The blue and red solid lines represent the fitting of this partitioning tracer model for K = 2.1 and K = 4, respectively. Mass-transfer Damköhler number in the microscopic zone is $Da_{m,1}$ = 300 and $Da_{m,2}$ = 600 for the fast and slow layer, respectively. The macroscopic mass-transfer Damköhler number is Da_M = 0.45. The fitting results give remaining oil saturation of $S_{o,1}$ = 0.17 and $S_{o,2}$ = 0.25 in the microscopic zone for the fast and slow layers, respectively. The fitted microscopic Damköhler number of $Da_m \gg 1$ indicates that the equilibrium assumption for partitioning tracer is valid for microscopic oil. Such behavior is anticipated for finely dispersed oil globules as there is enough specific oil-brine interfacial area to reach equilibrium. Additionally, the volumetric flow in the microscopic rock matrix is slow, and there is enough time for partitioning equilibrium to occur. On the other hand, the macroscopic Damköhler number

(Da_M = 0.45) indicates the non-equilibrium partitioning process in the macroscopic oil clusters, i.e. slow diffusion of the partitioning tracer into the oil blobs or patches. The distinction between microscopic and macroscopic oil saturation explains the fast arrival of the partitioning tracer of the fast layer, and the slow spread of the data after the peak. The partitioning tracer mass recovery of the proposed model is ~55% for both layers - higher than the expected mass recovery from the conservative tracer (~46%) for both layers. We observe that the proposed model overpredicts the partitioning tracer mass recovery while the equilibrium and non-equilibrium partitioning models underpredict the partitioning tracer mass recovery.

While the proposed model does not lead to a dramatic improvement in mass balance convergence for the partitioning tracer, it introduces several important advances over earlier formulations. First, it provides greater control over the recovered tracer mass. Previous models could not account for observed mass balance discrepancies in a physically consistent way. In contrast, the new model allows for such control and only requires parameter tuning to achieve closer agreement with measured data. Second, it can access information about oil saturation in the secondary layer — a capability absent in all prior models. This feature is particularly valuable, as it enables estimation of remaining oil saturation even when tracer datasets are short or incomplete.

Conclusions

We introduced and evaluated a new model for partitioning tracer transport in complex reservoir systems, applicable to both carbonate and sandstone formations. The model includes non-equilibrium partitioning in a dual-layer framework, representing microscopic and macroscopic oil regions, with partitioning tracer transport permitted between layers through interconnected oil phase. We applied this formulation to published data from a partitioning tracer test conducted in a multilayered carbonate reservoir and compared the results with prior interpretations. Overall, our findings are consistent with earlier studies. However, the proposed model offers greater flexibility for tuning mass balance convergence. More importantly, it enables estimation of remaining oil in the previously inaccessible secondary layer of the reservoir. Such a model is a promising tool to study complex naturally fractured reservoirs, such as carbonates, with dual-porosity and dual-permeability systems.

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